



Power and Centrality in Social Exchange

Where Centrality Breaks: Positive Sum vs. Zero-Sum Relations

The Social Exchange Perspective

All human interactions are “**exchanges**” that are social by nature

- Market exchange of goods/services
- Non-market exchange: gift, favors, advice, respect, emotion, invitation, etc.
- Interdependence (needs/wants) drives social interaction in the form of exchange

Trust is the basis of all exchanges (and social interactions)

- “How can I be sure that they won’t cheat?”
- Market vs. non-market exchanges use different mechanisms to solve the problem of trust
- **What** are they?

The Social Exchange Perspective

Market exchange:

- Terms of exchange: Negotiation
- Time frame: Immediate (spot market)
- Enforcement: Institutional sanctions, formal law
- Problem of trust: solved by central enforcers (state)
- Zero-sum: A's profit is B's cost

Non-market exchange:

- Terms of exchange: Reciprocity
- Time frame: Unspecified
- Enforcement: Social pressure, norms
- Problem of trust: solved by the decentralized collective (reputation, ostracism)
- Not clearly zero-sum: A's profit does not directly mean B's cost

The Social Exchange Perspective

Social ties we use to construct networks and the network measures we apply implicitly assume social interactions to resemble non-market exchange

- Reciprocity
- Social pressure discourages norm violation (trust from triadic closure)
- Power and influence grows with having more exchange partners (centrality)
- Not clearly zero-sum: A's social support to B can be reciprocated at a later time in-kind or with different resources (e.g., labor, status, loyalty)

Positive vs. Negative Connections

Network measures cannot be blindly applied to any network

Example: The Interdependence between ties
one exchange relation is contingent on the (non)exchange in a neighboring relation

- **positive connections**: Flow of resources from $B \rightarrow A \rightarrow C$. C can receive resources from A only if B transfers them to A.
- **negative connections**: zero-sum relations. A's exchange with B implies that A does not need to exchange with C i.e., B's gain is C's loss

Positive vs. Negative Connections

Positive
Connections

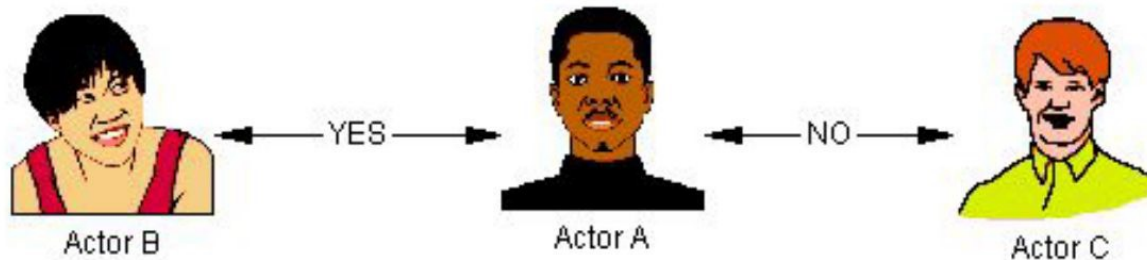
Positive sum relation



Example: supply chain

Negative
Connections

Zero-sum relation



Example: One buyer and
two competing sellers

Positive vs. Negative Connections

Degree centrality predicts “power” in networks of positive connections

Question: Does centrality predict power in networks of negative connections (i.e., zero-sum relations)? **Why?**

Power-Dependence Theory

For negative connections:

- If B depends on A more than A depends on B $\rightarrow$ A has that much more power over B ([Emerson, 1962](#))

$$P_{AB} = D_{BA}$$

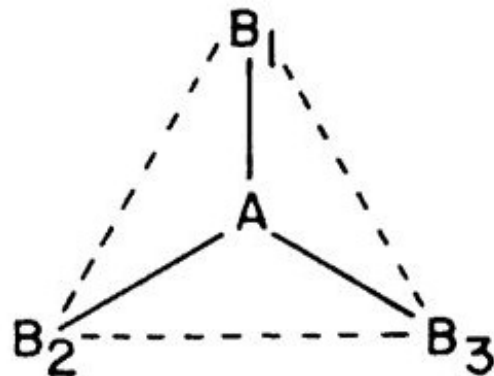
- These dependencies (hence power) stem from positions in the exchange network

So, does centrality and power-dependence logic make the same predictions about powerful positions in negatively connected networks?

Centrality in Negatively Connected Exchange Networks

Exchange network experiment by Cook et al. ([1983](#))

**1(a) 4 person network
(two positions)**



Lines: Exchange opportunities

- **Solid lines:** Two people negotiate how to split \$24 (larger pot)
- **Dashed lines:** Negotiate how to split \$8 (smaller pot)

-

Alphabets: Exchange positions

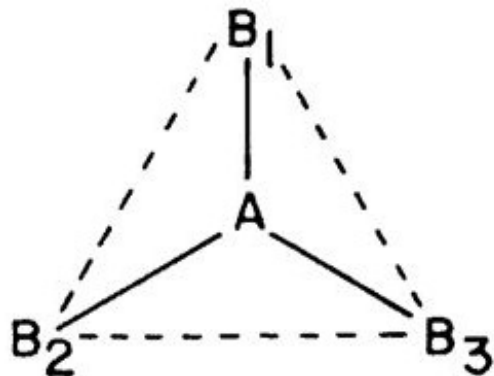
- Same alphabet positions (e.g., B1 and B2) are identical

Local knowledge: Participants do not know the exchange network (only the ties that they have)

Centrality in Negatively Connected Exchange Networks

Exchange network experiment by Cook et al. ([1983](#))

**1(a) 4 person network
(two positions)**



Network 1(a): A can exchange with only one among B1, B2, and B3 in one round

A and a partner in position B can negotiate how to split \$24 (solid line)

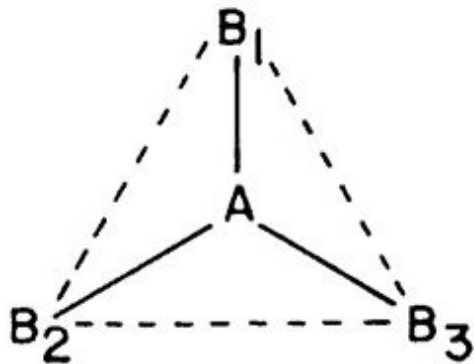
B1 and B2 can negotiate how to split \$8 (dashed line)

Negatively connected: In a round, If A chooses B1 as partner, then B2 and B3 cannot exchange with A

Predictions of Power According to Centrality

Exchange network experiment by Cook et al. ([1983](#))

**1(a) 4 person network
(two positions)**



Power: Measured by the total points that an occupant of a position earns through multiple rounds of exchanges

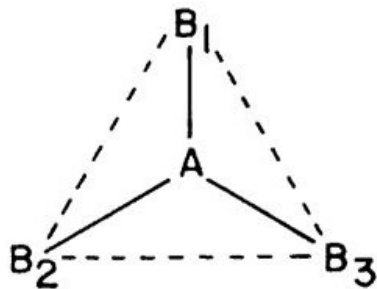
Centrality: A is the most central in terms of **weighted** degree, closeness, and betweenness centrality

Therefore, centrality predicts that power should be $A > B_1 = B_2 = B_3$

Predictions of Power According to Power-Dependence Theory

Power-dependence theory Prediction

I(a) 4 person network
(two positions)



If B1 exchanges with B2 or B3 → \$4

If B1 exchanges with A → any point above \$4 is more beneficial

If B1 suggests to A for an equal split (\$12:\$12)

B2 and B3 are in the same situation as B1

They will offer equal split to A

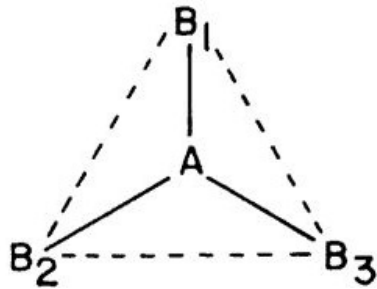
B1, B2, and B3 all **depend** on A to get a better outcome than \$4

In turn, A can bargain with all three for a better deal

Q: At equilibrium, what is the maximum that A will likely get?

Predictions of Power According to Power-Dependence Theory

I(a) 4 person network
(two positions)



At equilibrium: A's expected payoff: \$20, B's expected payoff: \$4

B position is dependent on A position to maximize payoff

→ **A's power over B is equal to B's dependence on A**

Power: $A > B1 = B2 = B3$

(Same as centrality)

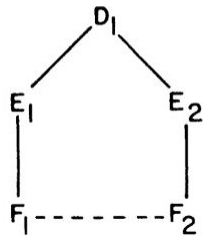
Predictions of Power According to Centrality

Exchange network experiment by Cook et al. ([1983](#))

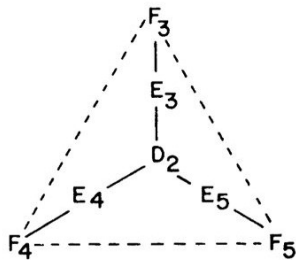
D is the most central in terms of (weighted) **closeness** and **betweenness** centrality for all configurations

Power: $D > E > F$

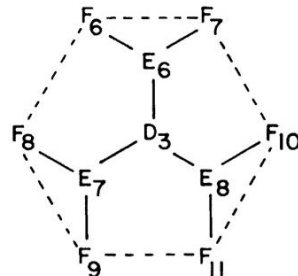
1(c) 5 person network
(three positions)



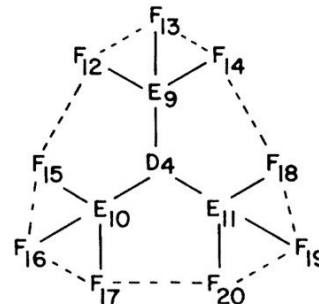
1(d) 7 person network
(three positions)



1(e) 10 person network
(three positions)



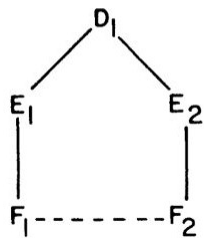
1(f) 13 person network
(three positions)



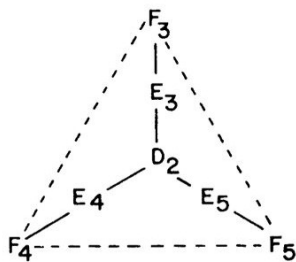
Predictions of Power According to Centrality

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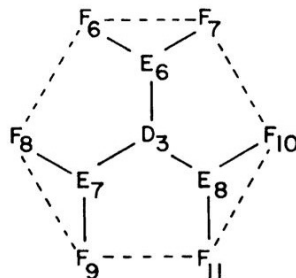
1(c) 5 person network
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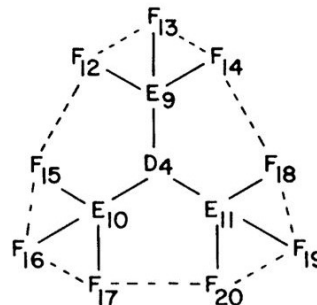
1(d) 7 person network
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1(e) 10 person network
(three positions)



1(f) 13 person network
(three positions)



D is the most central in terms of (weighted) **closeness** and **betweenness** centrality for all configurations

Power: $D > E > F$

D and **E** have same **degree** centrality in 1c and 1e

Power: $D = E > F$

D has larger degree than **E** in 1d

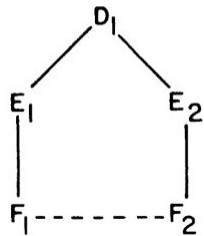
Power: $D > E > F$

E has larger degree than **F** in 1f

Power: $E > D > F$

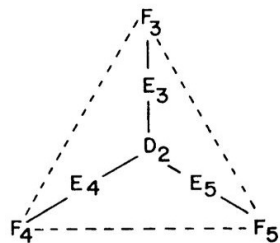
Predictions of Power According to Power-Dependence Theory

1(c) 5 person network
(three positions)

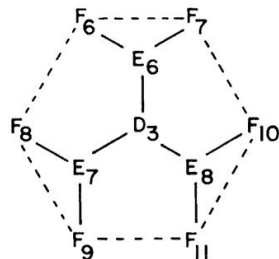


Q: Who is the most powerful according to power-dependence theory?

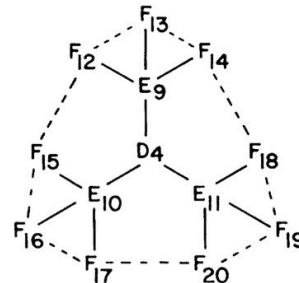
1(d) 7 person network
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1(e) 10 person network
(three positions)

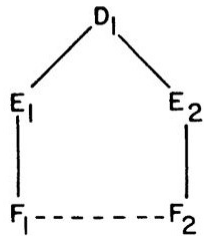


1(f) 13 person network
(three positions)

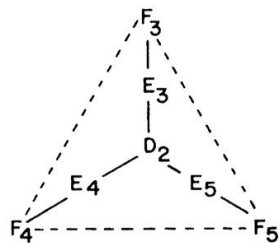


Predictions of Power According to Power-Dependence Theory

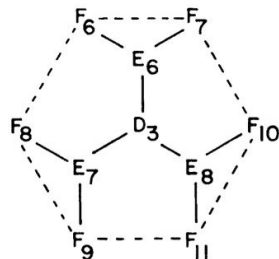
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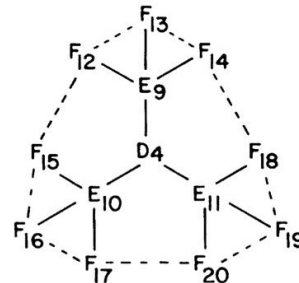
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1 (e) 10 person network
(three positions)



1 (f) 13 person network
(three positions)



Q: Who is the most powerful according to power-dependence theory?

F is dependent on E for higher payoff

→ E can ask for \$20 to F

→ E can also ask D for a “price match” (\$20)

D cannot earn more than \$4 because each E can exchange with their F for equivalent payoff

D's and F's expected payoffs will be \$4

Power: $E > D = F$

Experimental Evidence

Experiment designed with network 1c

Recruited 100 university students

27 transaction rounds

Negotiate with connected partners each round

Only one transaction per round per person

- negatively connected

Transactions are not revealed to others

1(c) 5 person network
(three positions)

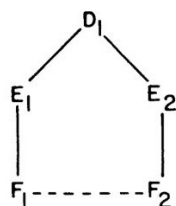


TABLE 1

EXPERIMENTAL RESULTS:
MEAN PROFIT OF PERSON E PER EXCHANGE WITH D AND WITH F
in NETWORK 1c BY EXCHANGE INCENTIVE AND TRIAL BLOCK

EXCHANGE INCENTIVE AND EXCHANGE PARTNER	TRIAL BLOCKS		
	1	2	3
Low:			
D	13.80 (4.13)	12.69 (4.26)	13.32 (4.25)
F	13.27 (3.10)	14.78* (2.77)	15.44** (2.96)
High:			
D	12.90 (3.71)	13.72 (4.40)	17.19** (5.26)
F	15.52** (2.38)	16.66** (2.10)	16.91** (2.46)
Combined:			
D	13.35 (3.95)	13.21 (4.36)	15.26** (5.16)
F	14.40** (2.99)	15.72** (2.63)	16.18** (2.82)

NOTE —The profit obtained by D and F in negotiations with E can be obtained by subtracting the values in this table (E's profit) from 24. Standard deviations are in parentheses

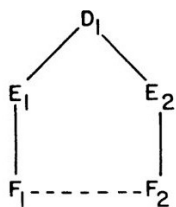
* Significantly greater than 12 ($P < .05$)

** Significantly greater than 12 ($P < .01$)

Experimental Evidence

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 Recruited 100 university students
 27 transaction rounds
 Negotiate with connected partners each round
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 - negatively connected
 Transactions are not revealed to others

1(c) 5 person network
 (three positions)



Why was E's power realized
 clearly in the last 9 rounds
 (Block 3)?

Why didn't E reach theoretical
 maximum (20 points)?

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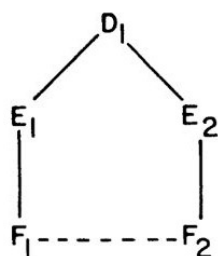
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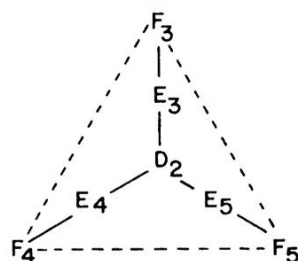
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Simulation Evidence

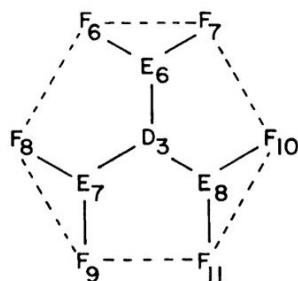
1 (c) 5 person network
(three positions)



1 (d) 7 person network
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1 (e) 10 person network
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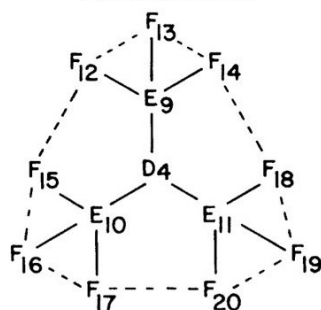


TABLE 2

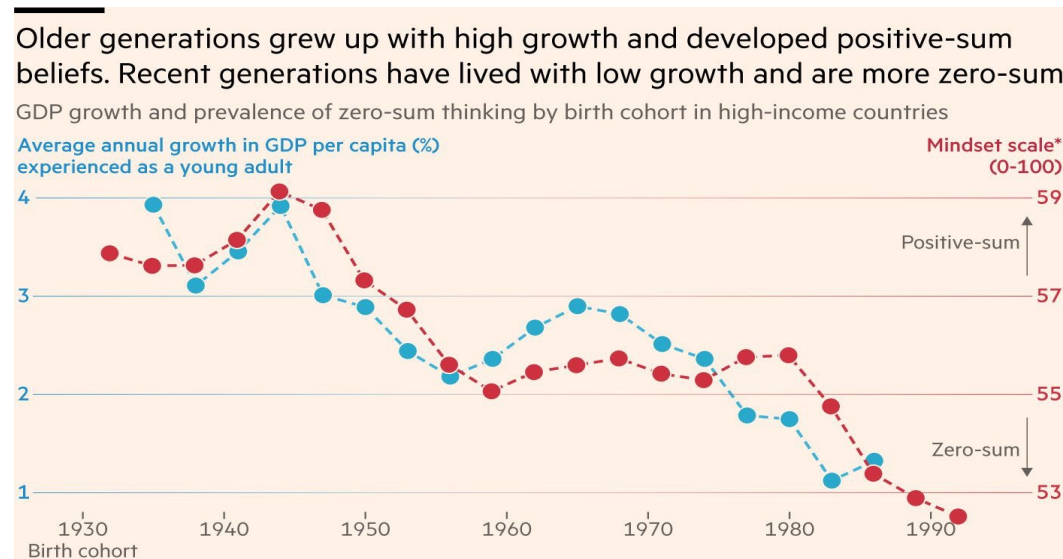
SIMULATION RESULTS:
MEAN PROFIT OF THE POWERFUL (E) PER "EXCHANGE" WITH D AND WITH F IN
FOUR REPLICATIONS VARYING NETWORK SIZE

FIGURE PART, SIZE OF NETWORK, AND E'S EXCHANGE PARTNER	TRIAL BLOCKS					
	1	2	3	4	5	6
1c, 5 actor:						
D	12.79a (2.57)	15.36 (2.76)	17.15 (2.98)	18.60 (2.11)	19.31 (1.67)	19.55 (1.44)
F	14.71 (2.64)	16.33 (2.76)	17.83 (2.56)	19.08 (1.86)	19.86 (1.42)	19.91a (1.26)
1d, 7 actor:						
D	10.56b (3.02)	13.03 (3.07)	15.00 (2.81)	16.54 (2.32)	17.64 (1.68)	18.47 (1.43)
F	14.33 (2.95)	15.31 (2.34)	16.64 (2.40)	17.79 (1.93)	18.66 (1.38)	19.06 (.97)
1e, 10 actor:						
D	13.99 (2.89)	17.69 (2.37)	19.65 (1.07)	20.06 (.58)	20.11 (.55)	20.11 (.54)
F	16.35 (2.16)	18.68 (1.35)	19.86 (.59)	20.11 (.42)	20.22 (.43)	20.01 (.39)
1f, 13 actor:						
D	14.50a (3.18)	19.56c (2.10)	20.42 (.86)	20.63 (.66)	20.50 (.64)	20.43 (.55)
F	17.18 (1.69)	20.06 (.74)	20.67 (.57)	20.87 (.59)	20.69 (.52)	20.58 (.49)

NOTE.—These values represent the average profit E obtained in "exchanges" with D and F, with 24 units of profit available for each "exchange"; therefore D's and F's average profit equals $24 - E$'s profit in each case. Each trial block contained nine trials. Cell values are based on the simulation of 50 groups; in an occasional group, however, E did not complete an "exchange" in a given trial block. Cell means labeled "a" are based on 49 groups, that labeled "b" has 47 groups, that labeled "c" has 42 groups per cell; all others have 50 groups per cell. Standard deviations are in parentheses.

Zero-Sum Relations

Centrality does not accurately capture power in networks of zero-sum relations
Then, centrality might not predict power in a society where people believe social life is zero-sum



*100 = "Wealth can grow so there's enough for everyone"; 0 = "People can only get rich at the expense of others"

Sources: FT analysis of World Values Survey; Maddison Project database

Based on *Zero-Sum Thinking and the Roots of US Political Divides* (Chinoy et al., 2023)

FT graphic by John Burn-Murdoch / @jburnmurdoch

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Summary

Different centrality measures for different aspects of power

Ask if centrality is the right way to think about power, given the nature of the tie (positive vs. negative connections)

Centrality does not quantify power accurately in negatively connected exchange networks

Power-dependence theory gives better prediction

Bonacich power centrality modifies eigenvector centrality to measure centrality in negative connections

Creative variation: Structural fragility